Hieu-Thi Luong

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AI researcher with 8 years of experience in speech processing, deepfake detection, and audio generative models. Developed state-of-the-art voice cloning systems, large-scale datasets, and patented technologies. Published in top-tier conferences and journals; skilled in signal processing, machine learning, and cross-national research collaboration.

Skills

Programming Languages: Python, Javascript, C/C++.
Machine Learning & Data Analysis: pytorch, librosa, numpy, pandas, matplotlib, espnet, kaldi.
AI & Speech Processing: Generative AI, Text-to-Speech (TTS), Automatic Speech Recognition (ASR), Deep Fake Detection
Others: Academic paper writing, Experimental design and implementation, Project management.
Languages: Vietnamese (native), English (proficient, C2), Japanese (beginner, A2).

Experience

Research Fellow, Nanyang Technological University – Singapore, Singapore — Jan 2024 – Present

- Developed an extensive 130-hour dataset of content-driven fake speech, encompassing realistic full and partial fake utterances, leveraging state-of-the-art voice cloning systems and signal processing techniques.
- Designed and implemented a pioneering fake speech creation tool, achieving a 10x productivity gain in data creation while substantially reducing reliance on large-scale data creation teams.
- Reimplemented and applied advanced signal processing technique to synthesize and manipulate Room Impulse Responses (RIRs), improving deepfake detection robustness and simulating realistic reverberant conditions.
- Managed multinational research collaborations, coordinating global teams to meet project milestones and generated high-impact results including academic papers, dataset and demonstration systems.

Postdoctoral Researcher, National Institute of Informatics – Tokyo, Japan — Oct 2020 – Mar 2023

- Developed innovative methods for audio/sound generation, specializing in human vocalizations, using remarkably small sound samples of under 60 seconds.
- Pioneered a new TTS/VC hybrid Voice Cloning technology, requiring less than 5 minutes of target speech, yielding one granted patent and advancing the state-of-the-art in speech synthesis.
- Applied Signal Processing and Speech Processing toolkits to tackle diverse research problems and challenges.

Research Assistant, National Institute of Informatics – Tokyo, Japan — Nov 2017 – Aug 2020

- Developed a novel Machine Learning methodology for multi-lingual and cross-lingual generative AI models, enabling bootstrapping of low-resource language systems and advancing global speech technology accessibility.
- Architected innovative Deep Learning models for multi-speaker Text-to-Speech and Voice Conversion systems, reducing the required data from 10 hours from a single speaker to 15 minutes from multiple speakers.
- Engineered and implemented high-quality Japanese Text-to-Speech models, addressing key technical challenges such as data imbalance and unique linguistic features like pitch accent.

Research Assistant, VNUHCM - University of Science – Ho Chi Minh City, Vietnam — Sep 2014 – July 2017

- Developed and published VIVOS, the first open-license Vietnamese Speech Corpus, accelerating R&D breakthroughs in Vietnamese Speech Processing Systems.
- Established state-of-the-art Automatic Speech Recognition (ASR) and Text-to-Speech baselines for the Vietnamese language, tackling its unique tonal characteristics and achieving a 28.94% relative error reduction.
- Supported lead instructors in instructional planning and student support for the Speech Processing course.

Education

The Graduate University for Advanced Studies, SOKENDAI – PhD in Multidisciplinary Science - Informatics — 2017 – 2020
VNUHCM - University of Science – MS in Computer Science — 2014 – 2016
VNUHCM - University of Science – BS in Information Technology — 2010 – 2014

Side Projects

ABC Notation Online Editor abc.hieuthi.com

- Free and online tool to compose music sheet using ABC Notation format

The Open-Source Joplin Plugin Development Project note-taking-joplin

- Multiple open-source plugins developed for Joplin to enhance the note-taking app user experience.

The Vietnamese technology, educational, lifestyle blog - ⅓ espresso 1p3espresso.com

- A self-hosted Vietnamese blog on technology, engineering, and education, sharing knowledge and experiences.

Patents

- JP2020027168A, Hieu-Thi Luong, and Junichi Yamagishi. "Learning device, learning method, voice synthesis device, voice synthesis method and program." Issued Jul 29, 2022.

Honors & Awards

- Singapore Academies South-East Asia Fellowship (SASEAF) for a 2-year Postdoctoral Research Fellowship (January 2024 - December 2025)
- Japanese Government (Monbukagakusho: MEXT) Scholarship for 3-year PhD Research (October 2017 - September 2020)

Publications (Selected)

<u>Luong, H. T.</u>, Li, H., Zhang, L., Lee, K. A., & Chng, E. S. (2025, April). **LlamaPartialSpoof: An LLM-Driven Fake Speech Dataset Simulating Disinformation Generation**. 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). IEEE.

<u>Luong, H. T.</u>, Truong, D. T., Lee, K. A., & Chng, E. S. (2024, December). **Room Impulse Responses help attackers to evade Deep Fake Detection**. 2024 IEEE Spoken Language Technology Workshop (SLT). IEEE.

<u>Luong, H. T.</u>, & Yamagishi, J. (2023, June). **Controlling Multi-Class Human Vocalization Generation via a Simple Segment-based Labeling Scheme**. Interspeech Conference 2023 (pp. 4379-4383). International Speech Communication Association.

<u>Luong, H. T.</u>, & Yamagishi, J. (2021, October). **Laughnet: synthesizing laughter utterances from waveform silhouettes and a single laughter example**. arXiv preprint arXiv:2110.04946.

<u>Luong, H. T.</u>, & Yamagishi, J. (2020, October). **Nautilus: a versatile voice cloning system**. IEEE/ACM Transactions on Audio, Speech, and Language Processing, 28, 2967-2981.

<u>Luong, H. T.</u>, & Yamagishi, J. (2019, December). **Bootstrapping non-parallel voice conversion from speaker-adaptive text-to-speech**. 2019 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU) (pp. 200-207). IEEE.

<u>Luong, H. T.</u>, Wang, X., Yamagishi, J., & Nishizawa, N. (2019, September). **Training multi-speaker neural text-to-speech systems using speaker-imbalanced speech corpora**. Interspeech Conference 2019 (pp. 1303-1307). International Speech Communication Association.

<u>Luong, H. T.</u>, & Yamagishi, J. (2018, September). **Multimodal Speech Synthesis Architecture for Unsupervised Speaker Adaptation**. Interspeech Conference 2018 (pp. 2494-2498). International Speech Communication Association.

<u>Luong, H. T.</u>, Wang, X., Yamagishi, J., & Nishizawa, N. (2018, September). **Investigating Accuracy of Pitch-accent Annotations in Neural Network-based Speech Synthesis and Denoising Effects**. Interspeech Conference 2018 (pp. 37-41). International Speech Communication Association.

<u>Luong, H. T.</u>, Takaki, S., Henter, G. E., & Yamagishi, J. (2017, March). **Adapting and controlling DNN-based speech synthesis using input codes**. 2017 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) (pp. 4905-4909). IEEE.

<u>Luong, H. T.</u>, & Vu, H. Q. (2016, December). **A non-expert Kaldi recipe for Vietnamese speech recognition system**. Proceedings of the Third International Workshop on Worldwide Language Service Infrastructure and Second Workshop on Open Infrastructures and Analysis Frameworks for Human Language Technologies (WLSI/OIAF4HLT2016) (pp. 51-55).